



AXOLOTL · AMBYSTOMA MEXICANUM

Cold water is the whole job

A complete, fact-checked printable manual: cold-water tank setup, cooling, cycling, water chemistry, feeding by age, handling, health triage, the law in every state, and the routines that keep an axolotl thriving for its full lifespan.

INSIDE

- ✓ Cooling a tank without a chiller, and when you need one
- ✓ Fishless cycling, step by step, and what to do if ammonia shows up
- ✓ Health triage: gills, fungus, impaction and floating
- ✓ Owner logs, pet-sitter sheet and a heat wave plan

45 pages

Intermediate

Print or tablet

Edition 3.0 · Oct 2026



CONTENTS

What's inside

Every page is written to be used, not just read once.

GETTING STARTED

How to Use This Package	3
-------------------------	---

SECTION 01 · PROFILE

Quick Profile & Cost Overview	4
The Species, Morphs, Size & Lifespan	5

SECTION 02 · HOUSING & ENVIRONMENT

Tank Size, Type & Where It Goes	6
Temperature: The Numbers That Matter	7
Chillers, Fans & Cooling Without One	8
Water Quality & the Nitrogen Cycle	9
Filtration, Flow & Lighting	10
Substrate, Hides & Furnishings	11
Water Changes, Cleaning & Hygiene	12

SECTION 03 · FEEDING

Diet & Feeding by Age	13
Worms, Pellets & Frozen Foods	14
The Never-Feed List & Supplements	15
Growth, Size & Body Condition	16

SECTION 04 · HANDLING & BEHAVIOR

Handling & Moving an Axolotl	17
Body Language & Normal Behavior	18
Enrichment & Common Mistakes	19

SECTION 05 · ARRIVAL & LIFE STAGES

Choosing an Axolotl, Quarantine & the First Vet Visit	20
Sexing, Pairs & Breeding	21

SECTION 06 · THE LAW

Is an Axolotl Legal Where You Live?	22
The Federal Rule, Virginia & Why the Bans Exist	23

SECTION 07 · HEALTH & COMMON ISSUES

Red Flags & Finding a Vet	24
Heat Stress & Reading the Gills	25
Fungal & Bacterial Infection	26
Impaction, Floating & Waste	27
Ammonia Burns, Parasites, Eyes & Obesity	28
Tubbing, Cooling & Salt Baths	29

SECTION 08 · QUICK REFERENCE

Setup Checklist & Targets	30
Emergency & Quick Targets Card	31

SECTION 09 · OWNER TOOLS

Budget & Shopping List	32
First 30 Days	33
Symptom Quick Reference	34
Daily, Weekly & Seasonal Routine	35
Power Outages, Heat Waves, Travel & Transport	36
Pet-Sitter Sheet	37
Owner Log	38
Equipment & Vet Log	39
Enrichment Checklist & Log	40

REFERENCE

Glossary, A to L	41
Glossary, M to Z	42
Sources	43
Where the Sources Disagree	44
Version History & About	45

START HERE

How to use this package

Three ways to get value from it, depending on where you are right now.

UNITS USED THROUGHOUT

Every temperature and size is given twice. Temperatures appear in degrees Fahrenheit (°F) with degrees Celsius (°C) beside them. Sizes appear in inches (in) with centimeters (cm) beside them. Tanks are measured in US gallons (gal), with liters (L) where a number matters. Water test readings are in parts per million (ppm): how many parts of a substance sit in a million parts of water. Small amounts are in milliliters (mL) and grams (g), and a salt bath's strength is in grams per liter (g/L). Prices are in United States dollars.

If you haven't set up yet

Check the law first, on page 22: an axolotl is legal without a permit in most of the United States, but it is banned, permit-only, conditional or unclear in fourteen places. Then read pages 4 to 12 in order before buying anything. Cooling is the decision that shapes the whole budget, and a new tank needs weeks of cycling before an axolotl goes in it. The Budget & Shopping List on page 32 lets you buy once.

If your axolotl is already home

Read the thermometer and test the water before you change anything else. Most axolotl problems that look like disease turn out to be a temperature or a water reading, and you cannot rule either out by looking at the animal. If anything seems off, go to the Red Flags page (page 24). If the water is already too warm, the steps that work in the next hour are on page 25.

Print these first

- | | |
|--|--|
| ☐ Page 30: Setup Checklist & Targets | ☐ Page 37: Pet-Sitter Sheet |
| ☐ Page 31: Emergency & Quick Targets Card | ☐ Page 38: Owner Log |
| ☐ Page 34: Symptom Quick Reference | ☐ Page 39: Equipment & Vet Log |
| ☐ Page 35: Daily, Weekly & Seasonal Routine | ☐ Page 40: Enrichment Checklist & Log |

WHY THIS PACKAGE EXISTS

Axolotl advice contradicts itself most on the two things that matter most: how cold the water has to be, and what is safe to put on the tank floor. This package gives numbers you can test against, cross-checked against the veterinary and research sources listed on page 43.

NOT A SUBSTITUTE FOR

An amphibian or aquatic veterinarian. This is a husbandry reference for day-to-day care. Anything urgent or health-related is always a vet call, not a page in this book. Vets who see axolotls are rarer than small-animal vets, so find one before you need one and write the number on the emergency card. Nothing on the law pages is legal advice either.

SECTION 01 · PROFILE

Quick profile

Everything at a glance before you read the detail.

Scientific name	<i>Ambystoma mexicanum</i> , a mole salamander and a close relative of the tiger salamander
Adult size	9 to 12 in (23 to 30 cm)
Captive lifespan	10 to 15 years with consistent care; 5 to 6 years in the wild
Native range	The lake and canal system of Xochimilco, on the southern edge of Mexico City
Water temperature	60 to 64°F (15.6 to 17.8°C). Chronic stress above 72°F (22°C), and possibly fatal from about 75°F (24°C). Never a heater
Tank minimum	A 20 gal long tank for one adult; a 40 gal breeder if you keep two
Water targets	Ammonia 0 ppm, nitrite 0 ppm, nitrate under 10 ppm, pH 7.4 to 7.6, in a fully cycled tank (page 9)
Diet type	Carnivore: earthworms or nightcrawlers as the staple, with a sinking pellet made for axolotls
Feeding	Juveniles about daily, adults every 2 to 3 days (page 13)
Sexual maturity	Males at about 10 months, females at 12 to 18 months
Handling	Only when necessary, and with a container rather than hands. A pet to watch, not to hold
Conservation status	Critically Endangered on the Red List kept by the IUCN (International Union for Conservation of Nature). Pet axolotls are captive-bred

Cost overview

Initial setup

\$30 to \$100 for a standard axolotl. \$120 to \$375 for the tank and equipment, or \$270 to \$775 with a chiller. Every line is on page 32.

Monthly ongoing

\$15 to \$45 for food, water conditioner, filter media and the electricity a chiller draws.

BUDGET FOR THE BAD MONTH TOO

An exam for an aquatic animal can cost more than one for a reptile or a bird, and tests, medication or surgery for a swallowed stone come on top. Keep a few hundred dollars set aside from the first week.

Fun facts worth knowing

- Axolotls regrow whole limbs, the tail, and parts of the eye, brain, spinal cord and heart, and the wound closes without a scar.
- They breathe four ways at once: through the skin, the lining of the mouth, the feathery gills, and simple lungs filled by gulping air at the surface.
- They have no eyelids, which is why dim, indirect light suits them.

The species, morphs, size & lifespan

What you are buying, how big it gets, and how long it stays.

A salamander that keeps its larval body

The axolotl is a salamander from the lake and canal system of Xochimilco, in Mexico City, that keeps its larval body for life: feathery external gills, a finned tail and a fully aquatic habit. The word for this is paedomorphosis, and it is not the same as staying a baby. An axolotl reaches full adult size and breeds normally. Nor is it strictly permanent: the tissues still respond to thyroid hormone, and a high enough dose in a laboratory can push one through metamorphosis into a land-dwelling adult. What is underactive is the hormone system that would trigger it. In a home tank that never happens, which is why a correct setup has no land area and no basking spot.

Wild and captive

In the wild the species is close to gone. Surveys of the Xochimilco canals counted about 6,000 axolotls per square kilometer in 1998 and 36 in 2014, a fall of 99 percent, driven by pollution and by introduced carp and tilapia that eat eggs and young. In laboratories and home tanks it is common: most laboratory axolotls descend from six of the 34 animals shipped from Mexico City to Paris in 1864.

Morphs, in plain terms

A morph is a color variety. Every morph needs exactly the same cold water, food and care.

MORPH	WHAT IT IS
Wild type	Dark olive and speckled, the color of every wild axolotl.
Leucistic	Mostly white with dark eyes: fewer pigment cells develop, and most fail to reach the skin. The pale pet-shop axolotl.
Albino	No black pigment at all, because the enzyme that makes it is broken. The gene came from a tiger salamander crossed into the laboratory colony.
Melanoid	Solid dark, with many black pigment cells and none of the reflective ones. Sometimes sold as melanistic.
GFP (green fluorescent protein)	Carries a jellyfish gene for a glowing protein, first put into axolotls in 2006 for research. Glows green under ultraviolet light, and can be combined with any color.
Mosaic and chimera	Built from two genetically different cell lines, so one body carries patches of two morphs. A mosaic does not breed more mosaics, which is the whole price premium.

The common colors come from single genes, mostly recessive, so two wild-type parents can both carry a hidden color. A dark morph hides early redness and fungus, so look closer under good light.

Size and lifespan

Most adults finish at 9 to 12 in (23 to 30 cm). An axolotl commonly lives 10 to 15 years in captivity and 5 to 6 in the wild. Plan for house moves, summers away and a decade or more of hot Augusts.

Tank size, type & where it goes

A bottom-dweller that walks more than it swims, in a tank that has to stay cold.

Size

A 20 gal long tank (about 76 L) is the minimum for one axolotl, and a 40 gal breeder (about 151 L) is the better answer if you keep two. Floor space matters more than height for this bottom-dwelling species, so a long, low tank beats a tall one of the same volume. Use a secure lid: an axolotl can climb out given the chance, and it dries out fast on the floor.

Volume also buys time. A large volume of water with a cycled filter holds its temperature and its chemistry between changes. Research colonies do the opposite, keeping each adult in about half a gallon of water that is replaced at least every other day, and that works because the water is changed before waste builds up. What fails is the middle ground: a small tank with no cycled filter and only weekly changes.

Where it goes

- The coolest room you have, away from windows, sunlight and heat-producing appliances. A cool room does part of a chiller's work for free.
- Somewhere you can reach the bottom with a siphon, since water changes are a weekly job.
- Near an outlet with room for a chiller or a fan, if the room runs warm in summer (page 8).

NEVER A HEATER

For almost every other aquarium animal, warmth is the fix. Here it is the hazard. If the tank came as a kit with a heater, take it out. An axolotl tank also needs no land area and no basking spot, because the animal never leaves the water.

One axolotl or two

SETUP	VERDICT	WHY
One adult alone	Safest	Axolotls are not social and do not need company. Housing separately is the safer default.
Two adults of similar size	Possible	Only with plenty of space, in a 40 gal breeder. Cohoused axolotls bite at each other's limbs and gills, most at feeding time.
Juveniles together	No	Juveniles are known for cannibalism. Raise young axolotls apart.
Mismatched sizes	No	Biting is worst when sizes differ, and anything that fits in the mouth gets swallowed.
Fish of any kind	No	Fish nip at the gills or take the food, and most want warmer water. Keep the tank species-only.

A male and a female together will breed, which is a much bigger commitment than it sounds (page 21).

Temperature: the numbers that matter

The single most important number in this package, and the one most often missed.

WATER READING	WHAT IT MEANS
60 to 64°F (15.6 to 17.8°C)	The target.
Above 72°F (22°C)	The ceiling. Extended time above it causes chronic stress and suppresses the immune system.
Above about 75°F (24°C)	Danger. Axolotls go off their food, turn sluggish, develop ascites (fluid swelling in the abdomen), float and cannot stay down, and pick up bacterial and fungal infections. The heat itself can be fatal.
41 to 59°F (5 to 15°C)	Not a target for a healthy tank. It is the range vets hold a sick axolotl in for weeks during treatment, which shows that cool water is not the danger warm water is.

Why warm water is so dangerous here

Two pieces of water chemistry stack. First, warm water holds less oxygen, at the moment a warmer animal is burning through more of it, and this one takes most of its oxygen straight from the water through its gills and skin. Second, warm water makes ammonia more toxic. Ammonia exists in two forms, and the un-ionized form is roughly 100 times more toxic than the other. Its share rises with temperature and with pH, so a small reading that a cold tank at pH 7.4 shrugs off becomes a real problem in a warm tank at pH 8. Heat does not just stress the axolotl directly. It turns every other small water problem into a bigger one.

MEASURE WHERE THE ANIMAL LIVES

Use a reliable thermometer in the water, not a strip on the outside glass, and read it every day. It is the reading most likely to change without warning, and catching drift early is the whole point.

Which plan your room needs

YOUR ROOM	WHAT TO PLAN FOR
Holds the low to mid 60s°F (around 16 to 18°C) all year	A thermometer, a fan on standby, and the tank away from windows and heat sources.
Cool most of the year, a few hot days	Air conditioning in that room and a fan across the water, ready before summer. Frozen bottles only as the emergency backup.
Above the mid 60s°F for weeks at a time	A chiller. Reaching for frozen bottles more than a couple of times a season is the sign it is time to buy one.

Whatever the method, change the water temperature slowly: no more than about 1°F an hour as a rule, never a drastic jump in either direction.

Chillers, fans & cooling without one

Ways to take heat out of a tank, the strongest first, and how to size a chiller.

METHOD	HOW IT WORKS AND WHERE IT FITS
Room air conditioning	Lowers the temperature the tank keeps trying to match. Nothing tank-side beats controlling the room.
Fan across the water surface	Evaporative cooling: moving air over open water pulls heat out of it. The best cheap active method. Needs an open top or a mesh lid, not glass.
Lights off, lid cracked open	Removes a heat source and lets warm, humid air escape. Free, and worth doing the moment a heat wave starts.
Less canister filter flow	A canister motor makes its own heat, and higher flow pushes it into the tank. A modest assist; never drop flow below what filtration needs.
Insulation and shade	Blocks heat from a sunny window or a warm room. Best for preventing a slow daytime climb.
Frozen water bottles	Sealed and floated, never loose ice. An emergency stopgap only: each bottle chills the water hard, then does nothing once it melts, and that swing is a stressor in its own right.
Aquarium chiller	The only method that holds a set temperature whatever the weather. \$150 to \$400, plus its electricity.

Run extra aeration alongside any of these. It does not cool the water, but warm water holds less oxygen, and more surface movement puts more in.

Sizing a chiller

A chiller is sold by the volume of water it can cool, and that rating comes with conditions. One maker rates its 150 L (about 40 gal) and 300 L (about 80 gal) units in an 86°F (30°C) room with the water starting at 82°F (28°C). The rated volume is the most a unit can handle under those test conditions, not a comfortable working load.

- Add the water in the filter and any sump to the tank's volume, and pick a unit rated well above that total.
- A 20 gal long sits comfortably inside a 40 gal rating. A 40 gal breeder is already at that unit's limit and belongs on the next size up.
- Go up another size if the room regularly runs hotter than the 86°F the rating assumes.
- Water is pumped through the chiller, so bring the return back through a spray bar or into a baffled corner. Axolotls are very sensitive to current.

THE RUNNING COST

The chiller's electricity is the cost most first-time owners do not budget for, and it varies most by climate and season. In a warm place, running one all summer adds up.

Water quality & the nitrogen cycle

Five numbers, a test kit, and the weeks of work that happen before the animal.

Axolotls excrete ammonia. In a cycled tank, one group of bacteria in the filter turns it into nitrite, and a second turns the nitrite into nitrate, which is far less toxic and leaves with water changes. A new filter needs weeks to grow those bacteria, so the tank goes through a full cycle before the axolotl goes in.

READING	TARGET	ACT WHEN	TEST
Temperature	60 to 64°F (15.6 to 17.8°C)	Above 72°F (22°C)	Daily
Ammonia	0 ppm	Any reading	2 to 3 times a week in a new tank, weekly once stable
Nitrite	0 ppm	Any reading	2 to 3 times a week in a new tank, weekly once stable
Nitrate	Under 10 ppm	Climbing past 10 ppm	Weekly
pH	7.4 to 7.6	Outside 6.5 to 8.0, or swinging	Weekly

Use a liquid test kit that reads ammonia, nitrite, nitrate and pH. Stability matters more than the exact pH, and the high end is where ammonia bites hardest.

Cycling the tank before the axolotl

1. Set the tank up complete, filter running and the water dechlorinated.
2. Dose pure ammonia, with no added soaps or dyes, to 3 ppm, inside the usual 2 to 4 ppm range.
3. Test daily or every other day, and dose back up to 3 ppm whenever it drops. Filter media from an established, healthy tank seeds the colony faster than anything sold in a bottle.
4. It is done when a full dose of ammonia reads 0 within 24 hours, nitrite reads 0, and nitrate is building. All three, not just one.

Most cycles take four to six weeks in warm water, and real tanks have taken anywhere from about three weeks to two months. A cold axolotl tank runs at the slow end: a drop of just 1°C (about 2°F) can cut the bacteria's work on ammonia by around 30 percent. Overdosing does not speed it up; it stalls the second half.

If ammonia or nitrite shows up

1. Change at least a third of the water with dechlorinated, temperature-matched water, and 50 percent or more if the reading is clearly climbing.
2. Skip or cut the next feeding, and remove any uneaten food.
3. Check the filter is running and that nobody rinsed the media in tap water.
4. Test again the next day, and repeat until both read 0.

The water itself

Tap water carries chlorine or chloramine, both deadly to skin and gills, so every drop gets a dechlorinator that handles both. Chloramine does not leave water that is left standing. Axolotls need mineralized water: never use straight RO (reverse osmosis) or distilled water without remineralizing it. No GH (general hardness, the dissolved minerals) or KH (carbonate hardness, which steadies pH) target for axolotls traces back to a published study, so this package prints none. Aquarium salt does not raise hardness.

Filtration, flow & lighting

Enough filter for a carnivore's waste, delivered gently, under a dim light.

Axolotls dislike current, and a filter built for an active fish will stress one. They are very sensitive to flow and can stop eating under a strong one. The answer is not a weaker filter. It is a properly sized filter whose output is broken up before it reaches the animal.

FILTER	HOW IT FITS AN AXOLOTL TANK
Sponge filter	The standard choice for axolotls: an air pump pushes water through a foam block, so the flow is gentle by design. Needs a separate air pump.
Canister filter	Holds the most media. Send the return through a spray bar, a bar of small holes that spreads the flow out, or into a baffled corner.

Signs the flow is too strong

- Gills curled forward toward the face. This can mean heat or water quality too, so check the thermometer and the test kit as well as the filter.
- Going off food. An axolotl can stop eating under a strong current.

A gentle current on one side of the tank, with a genuinely still corner on the other, lets the axolotl choose its own water movement.

Looking after the filter

The bacteria colony lives in the filter media, not the water. Rinse media in a bucket of water taken out of the tank during a change, never under the tap, since chlorine and chloramine kill the colony. Never replace all the media at once. A filter converts ammonia and nitrite into nitrate; it does not remove the nitrate, which is the job of water changes.

Lighting

Keep it dim. Axolotls have no eyelids and are sensitive to bright light, so floating plants or a hide give shade. Put the light on a timer for a 12-hour day, so the tank goes fully dark at night. In a heat wave, switch the light off altogether, since it adds heat.

A FILTER DOES NOT REPLACE WATER CHANGES

Nitrate builds up between changes whatever the filter does. Keep the weekly change even when every reading looks perfect.

Substrate, hides & furnishings

An axolotl gulps whatever sits by its food, so the tank floor is a safety decision.

Axolotls feed by suction: they open the mouth and take in everything in front of it, food and floor alike. Anything small enough to swallow and too big to pass is an impaction waiting to happen, and a blockage in the gut can need surgery.

FLOOR	VERDICT	DETAIL
Bare bottom	Safest	No impaction risk and the easiest to clean. The right answer for any juvenile under 5 in (13 cm).
Fine sand, grains under 1 mm (millimeter)	Good	Fine enough to pass if swallowed. For axolotls past 5 in, siphoned at every water change.
Gravel, small rocks, pebbles	Never	Swallowed with the food and too big to pass. The classic fatal mistake.

Wherever the floor is sand, feed on a bare patch, in a dish or from tongs, so the mouthful is food and nothing else.

Hides

Give at least 2 hides per adult, ideally 3 to 4 of different shapes and heights, and make one of them a tunnel, which tested as the item most worth adding (page 19). Hides are how an axolotl controls its light level, so they matter more for an animal that cannot close its eyes.

- Terracotta pots on their side with the drainage hole sealed.
- Wide PVC (polyvinyl chloride, a rigid plastic) pipe sections with every edge fully sanded.
- Smooth resin or ceramic caves.

Plants and decor

- Live or silk plants for cover. Java fern and anubias tied to decor rather than buried in the floor.
- Moss balls kept larger than the axolotl's head.
- Floating plants for dappled shade.
- Large stones and decor placed so they cannot topple or shift.

TWO TESTS FOR ANYTHING NEW

Run your hand over it: if it scratches you, it can tear an axolotl's skin. Then hold it up to the animal's head: anything small enough to swallow stays out of the tank. Every edge smooth, every piece stable, nothing loose and small.

Water changes, cleaning & hygiene

The one job that fixes more problems than every product on the shelf.

The weekly change

Change 20 to 30 percent of the water every week with dechlorinated water, which keeps nitrate down between tests. Test nitrate before the change. If it keeps climbing past 10 ppm, change more water or change more often, and check you are not overfeeding.

1. **Prepare the new water first.** Dechlorinate it, then bring it to within a couple of degrees of the tank, checked with the same thermometer you read the tank with. A bucket that feels fine to a hand can be ten degrees off.
2. **Siphon from the floor, not the surface.** Most of the waste settles on the bottom, so lift it off as the water comes out.
3. **Rinse filter media in the old water** while you have a bucket of it, never under the tap.
4. **Log it:** the date, the amount changed and the readings, on the owner log.

Every day

Remove uneaten food within 15 to 30 minutes of a meal. Leftover worm and pellet foul the water faster than anything else in the tank.

Cleaning tubs, tools and decor

JOB	HOW
Scrubbing a tub or bowl	Baking soda and salt mixed 2 to 1. Rinse well.
Disinfecting tools and housing	Household bleach at 30 mL per liter of water, with at least 30 minutes of contact. Rinse thoroughly in fresh water, then let it dry before use.
Quarantine	Separate tools, net and siphon for a quarantine tub, never shared with the main tank.
Filter media	Mechanical sponge or floss can be rinsed freely. Biological media barely ever, gently, in old tank water, and never all at once.

SKIN THAT ABSORBS WHAT IS IN THE WATER

An amphibian's skin and gills take in whatever is dissolved in the water, which is why chlorine that is harmless to people is lethal here. So every drop of new water is dechlorinated, top-ups included, and anything cleaned with bleach is rinsed thoroughly and dried before it goes back near the tank.

Diet & feeding by age

A carnivore that gulps its food whole and slows right down as it grows.

Axolotls eat invertebrate prey and need no plant matter at all. Earthworms or nightcrawlers are the staple, paired with a sinking pellet made for axolotls or carnivorous amphibians. Everything else is variety alongside those two.

STAGE	HOW OFTEN	WHAT
Hatchling, 0 to 3 months	2 to 3 times daily	Live food, such as daphnia and brine shrimp.
Juvenile, 3 to 6 months	About once a day	Cut worm pieces, small sinking pellets, frozen brine shrimp or daphnia.
Sub-adult, 6 to 12 months	Every 1 to 3 days	Cut worm and pellets, sized to the animal.
Adult, 12 months and over, past about 7.5 in (19 cm)	Every 2 to 3 days	Worm in head-sized pieces and pellets. Adults digest slowly in cool water and genuinely need the gap.

The 7.5 in mark is the point for the adult schedule, not full size: most adults go on to finish at 9 to 12 in (23 to 30 cm).

How much at each meal

- Size each meal to roughly the axolotl's head: what it takes in 5 to 10 minutes.
- Pull anything still uneaten after 15 to 30 minutes. Leftover worm and pellet foul the water.
- Feed from long, blunt tongs, on a bare patch or in a dish, so the mouthful is food and not floor, and so you can see exactly how much each animal ate.
- If you keep two, feed each one separately, since cohoused axolotls bite most at feeding time.

THE MOST COMMON FEEDING MISTAKE IS THE SCHEDULE

Feeding an adult on a juvenile's schedule. A grown axolotl fed daily becomes obese and loads the tank with waste. Obesity is a recognized problem in pet axolotls, and a measured portion at the adult interval is what prevents it.

APPETITE IS YOUR DAILY HEALTH SIGNAL

Poor appetite is one of the first signs of illness, and of warm or dirty water. Note every feeding response on the owner log. A refusal lasting 5 days or more is a vet call, after you have checked the thermometer and the test kit.

Worms, pellets & frozen foods

A short shopping list: one staple, one supplement, and some variety.

FOOD	TIER	NOTES
Earthworms and nightcrawlers	STAPLE	The gold standard: high in protein, well balanced, easy to digest and nutritionally complete. Cut into pieces, never fed whole.
Sinking pellets made for axolotls or carnivorous amphibians	STAPLE	The supplement to worms. Salmon-based or invertebrate-formulated pellets both work. Research colonies raise adults on sinking pellets above 30 percent crude protein alongside worms.
Frozen brine shrimp and daphnia	YOUNG, VARIETY	Good for hatchlings and juveniles, and as variety for adults. Not the whole diet on their own.
Frozen bloodworms	TREAT	A good occasional treat. Too thin to be the whole diet.

Worms: cut them up

Feed earthworms in portions rather than whole, even to a large adult: a piece sized to the head. That is also easier to count across the week than a whole nightcrawler of unknown length. For a juvenile, cut pieces are the only option.

Tongs and target feeding

Axolotls learn to take food from long, blunt feeding tongs, which is how research colonies feed them, and the tongs show exactly how much each animal ate. An axolotl notices food moving in the water through its lateral line (page 18), so a piece offered just in front of the face is found fast.

Making a meal take some finding

Dropping food in the same spot every time skips the hunting behavior. Scattering pieces so the axolotl has to find them is a free upgrade, as is a sand saucer: a shallow dish of sand under 1 mm with small food items hidden in it, which adds searching without loose sand across the floor. Rotate where you feed every few days. These change how the meal arrives, never how much the animal eats.

FEED ON A CLEAN PATCH

Because an axolotl gulps its food, it takes in whatever sits next to it. Feed on a bare patch, in a dish or from tongs, so the mouthful is worm and nothing else.

The never-feed list & supplements

A few foods that cause real harm, and the supplement step this species does not need.

FOOD	TIER	WHY
Feeder fish, live or frozen	NEVER	A real parasite risk, and many common feeder species carry an enzyme that causes a vitamin B deficiency over time.
Beef heart, chicken and other mammal meat	NEVER	Axolotls cannot properly digest mammalian fat and protein.
Adult mealworms, crickets, crayfish	NEVER	Hard-shelled feeders. The exoskeleton is difficult to digest and can cause internal injury.
Any food dropped onto gravel or small stones	NEVER	The axolotl gulps the stone with the food, and a stone that cannot pass blocks the gut.

The wrong food causes digestive and nutritional problems even without an impaction: feeder fish or mammal meat in place of worms and proper pellets is a slow harm, not just an emergency risk.

Supplements

None are generally needed. A varied diet of worms and formulated pellets covers this species' needs without added vitamins or minerals. There is no calcium dusting step and no vitamin schedule, which is one less thing to get wrong than with most exotic pets.

Feeding mistakes, ranked

MISTAKE	THE FIX
Feeding on loose small stones	The most consequential mistake. Bare bottom or fine sand, and feed from tongs or a dish.
An adult still fed daily	Every 2 to 3 days once past about 7.5 in and 12 months.
Feeder fish or mammal meat	Worms and a pellet made for axolotls.
Leaving food in the tank	Out within 15 to 30 minutes.
Feeding during a power outage	No food until the filter is back and stable. Uneaten food uses oxygen and rots in still water.

Growth, size & body condition

What changes as an axolotl grows, and the one body problem to watch for.

AGE	SIZE AND STAGE	WHAT CHANGES
Egg	Hatches in about 2 weeks at 70°F (21°C), sometimes 3 to 4	Tiny larvae that live on live food.
6 to 8 weeks	About 1 in (2.5 cm)	Still fed live food 2 to 3 times a day. Young axolotls kept together are cannibalistic, so each is raised apart.
3 to 6 months	Juvenile	About one meal a day, of cut worm, small pellets and frozen foods. Bare bottom while under 5 in (13 cm).
6 to 12 months	Sub-adult	Meals every 1 to 3 days. Males mature from about 10 months.
12 months and over	Adult schedule from about 7.5 in (19 cm)	Every 2 to 3 days. Females mature at 12 to 18 months.
Full grown	9 to 12 in (23 to 30 cm)	Lives 10 to 15 years in captivity.

Obesity

Obesity is a recognized problem in pet axolotls, and it comes from feeding an adult on a juvenile's schedule. A young axolotl eats daily; an adult needs food only every 2 to 3 days. The fix is the volume and frequency of meals: a head-sized portion at the adult interval.

Swelling is not fat

A swollen, fluid-filled body is ascites, not weight gain, and it points to heat stress or a bacterial infection. Bloating with floating and no waste points to an impaction. Both are health problems with their own pages (pages 26 and 27), not a feeding adjustment.

What to track

- Appetite at every meal, on the owner log. It changes before most other signs do.
- Gills: full and feathery is healthy; shrinking or curled forward is a water or heat problem, not a food one.

Handling & moving an axolotl

Why the answer is almost always no, and how to move one when you must.

Axolotls are not a hands-on pet, and that is about their bodies, not a preference. The skin is scaleless and sensitive, with no robust slime layer to fall back on the way a fish has. The feathery external gills do the actual work of breathing, so a snagged filament affects how the animal takes in oxygen, not just how it looks. And the skeleton is largely cartilage rather than solid bone. Handling risks skin and gill damage, real stress, and infection at any point of contact.

When a move is genuinely necessary

SITUATION	HOW
Tank maintenance, a rehouse, a health check	Guide the axolotl into a submerged container while it is still underwater, then lift the container. Keep it in water the whole time: air exposure and pressure are both real risks.
If a net is the only way	Avoid nets if you can, since gill filaments and toes snag in mesh. If nothing else works, use a soft, fine mesh net and have the axolotl in it for seconds, not minutes.
If it has to be touched	Wear clean, disposable nitrile gloves. The oils and residues on bare hands harm its skin.
A vet visit	A small, clear, watertight container filled with its own tank water, kept cool on the way (page 36).

At the vet

A vet visit is the most common reason an axolotl leaves its tank, and the container does most of the work. In a clear container the vet can look it over through the side without taking it out, and much of an axolotl exam happens that way. When a vet needs to restrain one for a closer look, a sample or a procedure, the standard is sedation in the water with an anesthetic drug, rather than holding a struggling animal by hand.

Bringing a new axolotl home

1. **Temperature first.** Float or stand the travel bag or container beside the destination water until a thermometer reads the same in both.
2. **Then chemistry, slowly.** Drip or add small amounts of the destination water to the container over time, so the chemistry shifts gradually. This is drip-matching, and it settles the animal with no handling at all.
3. **Then the move.** Guide it across in a submerged container. Know the targets in advance: 60 to 64°F (15.6 to 17.8°C) and pH 7.4 to 7.6.

If another axolotl already lives in the tank, the newcomer goes into its own quarantine tub instead, and the acclimation happens into that tub (page 20).

THE RELATIONSHIP IS REAL, IT IS JUST NOT TOUCH

If part of what you want from a pet is physical contact, an axolotl will not offer it, and pushing for it harms the animal. What it does offer is a lot to watch, and an animal that learns to take food from your tongs.

Body language & normal behavior

What a healthy axolotl does all day, and what its gills and posture tell you.

What normal looks like

Most of an axolotl's day is spent lying nearly motionless on the bottom. That is normal for the species, not a sign that the tank is boring or the animal is ill. It notices food through its lateral line, a row of pressure-sensing organs along the body, and will snap at a pellet dropped in front of it as it sinks. It breathes four ways: through the skin, the lining of the mouth, the gills, and simple lungs it fills by gulping air at the surface. So an occasional trip up for a gulp of air, then back down, is normal too.

NORMAL	CONCERNING
Long stretches of stillness on the bottom	Lethargy together with appetite loss
Full, feathery gills held out to the sides	Gills curled forward toward the face
An occasional gulp of air at the surface	Gulping air much more than usual
Resting on the floor of the tank	Floating at the surface and unable to stay down
Eating on schedule, taking food from tongs	Appetite loss lasting 5 days or more
Sitting in a hide or under plant cover	White cottony tufts, red patches, streaks or sores

Reading the signs

WHAT YOU SEE	WHAT IT USUALLY MEANS
Gills curled forward	Stress: heat, poor water, or too much current. Often well before appetite loss, which is why the gills are worth a look every day.
More air gulping than usual	The water is not delivering enough oxygen through the gills. Check the temperature and the surface movement.
Floating it cannot correct	One of the named heat signs above about 75°F (24°C), or bloating from an impaction.
Off food and sluggish	Often the first response to ammonia, nitrite or temperature. Test before anything else.

TEST FIRST, EVERY TIME

An axolotl often shows a water problem before a test does. Any change in the gills, appetite or floating means reading the thermometer and testing the water first, before reaching for anything else (page 24).

Enrichment & common mistakes

Not toys. Choice: somewhere to rest, cover that feels secure, and food that takes some finding.

What has been tested

Billingham's 2017 study gave laboratory axolotls rocks, plants, plastic shelters, old compact discs and tunnels, and recorded what they did. The animals nudged, rubbed against and moved the items, in one case shifting a disc into a position that made a shelter. Tunnels came out as the item most worth adding. Research colonies land on the same short list: plants, places to hide, controlled temperature and a set day length. A timer on a dim light, on the same schedule every day, costs nothing.

What to offer

- **Structure:** 3 to 4 hides of different shapes, one of them a tunnel, most of them kept stable and never moved.
- **A choice of current:** a gentle flow on one side from a baffled filter, and a genuinely still corner on the other.
- **Foraging:** tong feeding, scattered pieces, or a sand saucer, with the feeding spot moved every few days.
- **Novelty:** a mild rearrange of non-critical decor about once a month, or one new smooth item in for 5 to 7 days, then rotated out.
- **Shade:** floating plants or a dimmed light for the dappled shade this species is drawn to.

Frequent full overhauls remove the security the tank is supposed to give. Cohousing is not enrichment either: two axolotls bite at limbs and gills. Strong current and bright light are just stress.

Common mistakes, ranked by harm

MISTAKE	COST	THE FIX
Water running warm	Fatal	Plan cooling before summer, not during it. Never above 72°F (22°C).
Gravel or small stones	Fatal	Bare bottom, or fine sand under 1 mm.
No cycle before the axolotl	Severe	Cycle first; ammonia and nitrite burn from day one.
A heater in the tank	Severe	There is no case for one. Take it out.
Fish as tankmates	Severe	Species-only. Fish nip gills and take the food.
Two axolotls of different sizes	Injury	One axolotl as the default; two only with space and similar size.
Strong filter flow	Chronic	Sponge filter, or a spray bar on the return.
An adult fed daily	Chronic	Every 2 to 3 days.
Bright light	Chronic	Dim light on a timer, with shade.
Handling, or a net for routine moves	Injury	A submerged container. Gloves only if touch is unavoidable.
No lid	Escape	A secure lid. Axolotls can climb out.
Treating before testing	Lost days	Thermometer and test kit first.

Choosing an axolotl, quarantine & the first vet visit

What to look at before you pay, and the weeks before a newcomer meets an established tank.

Checks before you buy

CHECK	WHAT YOU WANT, AND WHY
The water	Feel it or ask its temperature. An axolotl kept above about 75°F (24°C) turns sluggish and open to infection, and carries that home.
The gills	Full and feathery, held out to the sides. Curled forward is a stress sign.
The skin	No white, cotton-like tufts (water mold, the most common fungal infection), and no red patches or open sores.
How it sits	Resting on the bottom. Not floating at the surface, and no swollen, fluid-filled belly.
Appetite	Ask to see it eat. Poor appetite is one of the first signs of illness.
The tank floor	Skip animals kept on gravel. One that has lived on pebbles may already have swallowed some.

Quarantine

If you already keep an axolotl, a new one never goes straight into the established tank. It spends 6 to 8 weeks in its own tank or tub first, typically adequate for a new amphibian. Zoos run 30 days as a minimum and prefer 60. Land closer to 8 weeks than to the shortest end, and restart the clock, not just pause it, if the animal shows any illness partway through.

- A second tank or a large tub with a secure, ventilated lid, kept bare or close to it so you can watch the animal and its waste.
- Its own net or scoop, siphon and cleaning tools, never shared with the main tank.
- A fecal test (a vet's microscope check of a stool sample for parasites) through an exotics vet at some point in the window.

WHY STRICTER THAN FOR A FISH

Two fungal diseases target amphibians. Bd (*Batrachochytrium dendrobatidis*) affects more than 200 species. Bsal (*Batrachochytrium salamandrivorans*) kills salamanders, axolotls included, and reached Europe through the pet trade. Over 750,000 salamanders were imported into the United States between 2010 and 2014 alone. Never release an axolotl or its tank water outdoors.

The first vet visit

Find an aquatic or exotic-amphibian vet before you need one: general small-animal clinics often are not equipped for this species. The find-a-vet directory of the Association of Reptile and Amphibian Veterinarians lists candidates; ask each practice directly whether it sees axolotls. Bring the animal in a container of its own tank water, plus a separate sample of tank water in a clean container and your latest test readings.

Sexing, pairs & breeding

How to tell males from females through the glass, and why a pair is a bigger commitment than it sounds.

Sexing, with no handling at all

FEATURE	MATURE MALE	MATURE FEMALE
Cloaca (the vent behind the back legs, used for waste and breeding)	Swollen: a visible bulge behind the back legs	Slimmer there, with no bulge
Nails	Dark	Not dark
Age at maturity	About 10 months	12 to 18 months

An axolotl younger than that cannot be sexed reliably by anyone, so a juvenile sold as a particular sex is a guess.

A pair will breed

A mature male and female kept together will breed. The female lays her eggs 12 to 20 hours after mating, spread over one or two days. In managed care a female can lay 200 to 1,500 eggs every 3 to 6 months. At 70°F (21°C) the eggs hatch in about 2 weeks, sometimes 3 to 4, into tiny larvae that live on live food, 2 to 3 times a day.

Raising them is the real commitment. Juveniles are cannibalistic and are raised apart, so a single clutch means hundreds of animals to house, feed live food and rehome, and they keep coming every few months.

IF YOU DO NOT WANT HUNDREDS OF LARVAE

Keep one axolotl, or two of the same sex, confirmed after maturity.

What sets off breeding

In one research vivarium, natural breeding followed a cool spell of 12 to 14°C (54 to 57°F) with about 11 hours of light a day. It is a reminder that the light cycle is part of the environment too.

No metamorphosis

An axolotl reaches maturity and breeds while still in its larval body, and it never leaves the water. The seasonal job in this species belongs to the keeper: getting the tank through summer heat, which is planned on the routine page.

Is an axolotl legal where you live?

Legal without a permit in most of the country, and banned or restricted in fourteen places.

This page covers all 50 states, the District of Columbia and New York City, which has its own rules: 52 jurisdictions. A pet axolotl is **legal with no permit in 38, banned in 6, permit-only in 4, legal on a condition in 3, and unclear in 1.**

WHERE	STATUS	THE RULE
California	Banned	California Code of Regulations, title 14, section 671 restricts the whole genus <i>Ambystoma</i> , and no permit type covers pet keeping.
New Jersey	Banned	Named in the state wildlife division's restricted species notice, which bars keeping or selling axolotls (New Jersey Administrative Code 7:25-4.10(a)). A neotenic salamander cannot be told apart from the larva of a protected native one by looking.
New Mexico	Banned	Left off the Director's Species Importation List (2023), so the default puts it in group IV, which the general public may not import. The rule governs bringing one into the state.
Wyoming	Banned	Chapter 69, section 8(a): no permit is issued for wildlife on the federal injurious wildlife list, which has included genus <i>Ambystoma</i> since 2025.
Alabama	Banned	Alabama Administrative Code r. 220-2-.26(1) bans possessing any amphibian on the federal injurious wildlife list, so the 2025 federal rule made it a ban.
District of Columbia	Banned	District of Columbia Code § 8-1808(j)(1) allows seven groups of household pet, and no amphibian fits any of them.
Maine	Permit	Not on the Unrestricted Species List, whose amphibians are frogs and toads only. The list's conservation footnote would also catch a Critically Endangered species.
Vermont	Permit, not for pets	Genus <i>Ambystoma</i> is on the Restricted Wild Animal List (July 2025), and permits are issued only for scientific or educational purposes.
Massachusetts	Permit, not for pets	Code of Massachusetts Regulations, title 321, section 9.01: a Critically Endangered species is not exempt, and the state says it will not issue a permit to keep a wild animal as a pet.
West Virginia	Permit	West Virginia Code § 20-2-13: live wildlife from out of state needs an importation permit from the director, and amphibians count, captive-bred or not. No application is published: ask in writing before the animal crosses the state line, and keep the reply.
Hawaii	Conditional	Named on the List of Conditionally Approved Animals: import and possession are allowed subject to Department of Agriculture conditions.
Rhode Island	Conditional	Section 3.7(H) of the state's exotic animal rules: kept indoors as a pet, in a way that prevents escape. Importing is limited to licensed pet shops and researchers.
Minnesota	Conditional	Minnesota Statutes § 97A.401, subdivision 8: a salamander from a permitted breeder, or held before 1 August 2021, may be kept as a pet. No breeder permit is visibly in place yet, so ask the state wildlife agency how to meet the condition.
Arkansas	Unclear	The unrestricted captive wildlife list has no amphibian section, so a non-native salamander is neither exempted nor addressed.

Legal with no permit: 38 jurisdictions

Alaska, Arizona, Colorado, Connecticut, Delaware, Florida, Georgia, Idaho, Illinois, Indiana, Iowa, Kansas, Kentucky, Louisiana, Maryland, Michigan, Mississippi, Missouri, Montana, Nebraska, Nevada, New Hampshire, New York State, New York City, North Carolina, North Dakota, Ohio, Oklahoma, Oregon, Pennsylvania, South Carolina, South Dakota, Tennessee, Texas, Utah, Virginia, Washington and Wisconsin.

In eleven of them, Colorado, Illinois, Kentucky, Maryland, Michigan, Montana, Nevada, North Carolina, Ohio, Oregon and Washington, the state lets cities or counties be stricter, so check your local ordinance (a city or county law) too.

The federal rule, Virginia & why the bans exist

One federal change that moved two states, and one ban that ended in 2021.

The 2025 federal rule

A federal rule effective January 10, 2025 added 16 more salamander genera to the list of injurious wildlife under the Lacey Act, the federal law that lets the government restrict harmful wildlife, bringing the total to 36 genera. All of them are tied to the risk of Bsal, the salamander-killing fungus, and genus *Ambystoma*, which includes the axolotl and the tiger salamanders, is among the new ones.

WHAT THE LISTING RESTRICTS

Importing axolotls into the United States, and shipping them to or from Hawaii, the District of Columbia, Puerto Rico or the US territories.

WHAT IT DOES NOT RESTRICT

Owning, breeding or selling an axolotl within the 49 continental states. The US Fish and Wildlife Service states that distinction directly in its own answers on the rule.

The twist is in two states. Wyoming and Alabama do not write their own lists for this; they bar whatever is on the federal injurious list. So for owners there, the January 2025 rule was the moment the animal became illegal to keep. And Hawaii's conditional yes sits beside a federal limit on shipping one there.

Virginia is legal, and has been since 2021

Virginia still shows up on ban lists across a surprising number of hobby sites. Since August 1, 2021 it has been legal to import, export, sell and possess axolotls there with no permit, after an amendment to the state's administrative code carved axolotls out of its general ban on selling salamanders. A list that shows Virginia as banned predates the change.

Why the bans exist

None of the restrictions protect axolotls. They protect native salamanders. The axolotl sits in genus *Ambystoma* with North America's mole and tiger salamanders, including California's own tiger salamander, which is federally threatened. Regulators weigh three risks: hybridization with native salamanders, an escaped population establishing itself, and disease, the chytrid fungi above all. California's label for the genus, a detrimental species, says so outright. New Jersey adds an enforcement reason: a gilled adult axolotl looks like the larva of a protected native salamander.

NEVER RELEASE ONE

An axolotl, its tank water or its plants never go into any pond, stream or drain. A captive amphibian carrying Bd or Bsal becomes a wild exposure the moment it is loose, which is the whole reason for the federal listing.

NOT LEGAL ADVICE

State legal status as of October 2026. Both law pages summarize state and federal rules, read on 1 October 2026, and are not legal advice. Rules change. Before you buy or move, check with your state wildlife agency and with your city or county.

Red flags & finding a vet

What to watch for, when to call, and what to have ready when you do.

SEE AN AQUATIC OR EXOTIC-AMPHIBIAN VET PROMPTLY FOR

Fungal growth that has not improved within 48 hours of correcting the water · suspected impaction: bloating, floating instead of resting on the bottom, no waste · red streaking, reddened skin or open wounds · a swollen, fluid-filled body · uncontrollable floating · appetite loss lasting 5 days or more.

Test before you treat

A large share of what goes wrong traces back to water that is too warm, and most of the rest to water quality. So the first two actions for any symptom are a thermometer reading and a full test of ammonia, nitrite, nitrate and pH. An axolotl often shows a water problem before a test does, and medication into a tank running at 75°F (24°C) fixes nothing. Correcting the water is not a delay in treatment: it is usually the largest part of it. Overfeeding and the wrong food are common triggers too.

Finding a vet before you need one

General small-animal clinics often are not equipped for this species, and warm water or a swallowed stone rarely waits for business hours. Use the find-a-vet directory of the Association of Reptile and Amphibian Veterinarians to find candidates, ask each practice directly whether it sees axolotls, and write the number on the emergency card. Some exotic hospitals book aquatic animals as a longer, two-part appointment, which costs more than a reptile or bird exam.

What to have ready when you call

- ☐ The water temperature, and how long it has been out of range
- ☐ Your latest ammonia, nitrite, nitrate and pH readings, with the date
- ☐ A sample of tank water in a clean container, to bring along
- ☐ Tank size, filter and floor type, and how long the tank has run
- ☐ What it eats, how often, and when it last ate
- ☐ When the signs started, and the owner log

GETTING IT THERE

A small, clear, watertight container of its own tank water, kept cool on the way. The vet can examine it through the side without taking it out.

Heat stress & reading the gills

The condition most axolotl health problems trace back to, and the early warning on the animal's own head.

Cause: water above 72°F (22°C) for any length of time suppresses the immune system; above about 75°F (24°C) the heat itself can be fatal. **Signs:** gill tips curled forward, appetite loss, floating, and pale patches on the skin. Above about 75°F, axolotls turn sluggish, float uncontrollably, develop ascites and pick up bacterial and fungal infections. **Response:** bring the temperature down, steadily, and test the water while you do.

The water is too warm, right now

1. Lights off, lid cracked open, and a fan blowing across the water surface.
2. Close the blinds, and run the room as cool as you can.
3. Float sealed bottles of frozen dechlorinated water, swapping them as they melt. Never loose ice: it shocks the water around each cube.
4. Lower the temperature no faster than about 1°F an hour, never in a drastic jump.
5. Check the filter output too: gills curled forward can mean too much current rather than heat.
6. Test ammonia and nitrite, since warm water makes the same ammonia reading more toxic.

AN AXOLOTL ALREADY IN CRISIS

For an animal in outright heat crisis, from long exposure above 75°F (24°C), placing it in a dish in the refrigerator is a recognized stopgap while the main tank is cooled. It is a short emergency measure, not routine care.

Reading the gills

WHAT THE GILLS LOOK LIKE	WHAT IT USUALLY MEANS
Full and feathery, held out to the sides	Healthy. Note what yours look like on a good day, as a baseline.
Curled or angled forward toward the face	Stress: heat first, then water quality or current. Often well before appetite loss.
Red, inflamed patches	Ammonia or nitrite burn. Test now (page 28).
White, cotton-like tufts	Fungal infection taking hold on stressed tissue.

WHY THE GILLS ARE SUCH A GOOD EARLY WARNING

They curl or angle forward under heat stress well before more obvious signs like appetite loss show up. A look every day, not just when something already seems wrong, buys days of notice.

Fungal & bacterial infection

The two infections that follow warm water and dirty water, and how to tell them apart.

Fungal infection

Signs: a white or gray, cotton-like growth on the gills or body, the water mold that is the most common fungal infection in the species. **Cause:** warm water, poor water quality, stress, or an existing injury that gave the fungus an opening. **Response:** correct the water quality and cool the tank. Salt baths can help, but the one published figure is a ceiling, not a dose, and the working dose belongs to an exotics vet (page 29). Severe cases need a vet-prescribed antifungal, and so does any case not improving within 48 hours of the water being right.

Bacterial infection

Signs: red streaks, reddened skin, open sores, and a swollen, fluid-filled body (ascites). **Cause:** most often *Aeromonas* or *Pseudomonas* bacteria, which take hold when the water runs warm or dirty. **Response:** a vet and prescription treatment. Water changes alone will not clear it. Caught early, it is treated with systemic antibiotics such as enrofloxacin or a trimethoprim-sulfonamide, and once advanced the outlook is poor, which is the reason to act on the first signs.

TELLING THEM APART	FUNGAL	BACTERIAL
What you see	Fluffy white or gray tufts standing off the skin or gills	Redness, streaks, sores, swelling, no fluff
Usual trigger	Warm or poor water, stress, an injury	Warm or dirty water
First response	Fix the water and cool; vet if no better in 48 hours	Fix the water and call a vet the same day

WHAT A VET MAY DO

During an antibiotic course for a bacterial infection, a vet may hold the axolotl at 41 to 59°F (5 to 15°C) for several weeks, because cooler water appears to help a sick axolotl. That is part of a treatment a vet is managing, not something to try alone. This is here so you can follow the plan, not so you can medicate at home.

PREVENTION REALLY IS THE TREATMENT

Both are opportunists. An axolotl in cold, clean, cycled water, with no injuries and no tankmate biting it, rarely gets either.

Impaction, floating & waste

The preventable emergency, and how to tell it from the harmless version of the same symptom.

Cause: swallowed gravel or small decor, straight from the way an axolotl feeds: it gulps whatever sits next to its food. **Signs:** bloating, floating instead of resting on the bottom, appetite loss, and no waste.

Response: a vet. Mild cases sometimes resolve with fasting and cooler water under a vet's guidance, but a blockage low in the gut can need surgery.

Floating: four presentations that look alike

WHAT YOU SEE	LIKELY CAUSE	WHAT TO DO
Rises for a gulp of air, then swims back down	Normal	Nothing. Axolotls breathe partly through simple lungs.
Floats, with curled gills, in a warm tank	Heat stress	Cool the water gradually and reassess (page 25).
Floats, bloated, with no waste	Impaction	Stop feeding and call a vet.
Floats with a swollen, fluid-filled body	Ascites: heat or a bacterial infection	Vet, promptly.

Any floating it cannot correct, whatever the cause, is a call to the vet rather than a wait-and-see.

Waste as a signal

A healthy axolotl that is eating passes waste. No waste after meals, especially with bloating, is one of the named signs of impaction. Waste and leftover food on the floor are also the main source of ammonia, so siphon them out at every water change.

Preventing it entirely

- Bare bottom for any axolotl under 5 in (13 cm), and fine sand under 1 mm at most for the rest. Never gravel or small rocks.
- Feed from tongs, on a bare patch or in a dish, so the mouthful is food and nothing else.
- Worm in head-sized pieces, and no hard-shelled feeders.
- Nothing small enough to swallow anywhere in the tank.

DO NOT WAIT IT OUT

If you know the axolotl swallowed a stone or a piece of decor and it has stopped eating, that is a vet call today. Fine sand or a bare floor prevents almost all of it.

Ammonia burns, parasites, eyes & obesity

The rest of the list, most of which is water chemistry or feeding wearing a different face.

Ammonia and nitrite burns

Signs: red, inflamed patches on the skin or gills. **Cause:** an uncycled or poorly maintained tank.
Response: test, then change at least a third of the water with dechlorinated, temperature-matched water, 50 percent or more if a reading is clearly climbing, and repeat until ammonia and nitrite both read 0 ppm. The real fix is a properly cycled tank, checked with a test kit before you rely on it.

CONDITION	SIGNS	CAUSE AND RESPONSE
Gut parasites	Found with a fecal test	Picked up from food or tankmates. <i>Hexamita</i> , <i>Opalina</i> , <i>Trichodina</i> and <i>Ichthyobodo</i> are among them. A fecal test at the vet, ideally during quarantine, and vet treatment if found. One more reason never to use feeder fish.
Eye problems	Changes in the eyes	Linked to prolonged high nitrate. Keep nitrate under 10 ppm with weekly changes, and see a vet if it persists.
Obesity	A heavy, overfed body	Feeding an adult on a juvenile's schedule. Adults eat every 2 to 3 days, in head-sized portions.
Appetite loss	No interest in food	Check temperature and water first; then look for impaction or infection. A vet visit at 5 days or more.
Bite wounds	Missing toes, gill filaments or limb tips	From a tankmate. Separate them: a tub isolates an injured animal without disturbing the tank. Watch the wound for white fluff or redness.

Regeneration is not a reason to relax

Axolotls regrow lost limbs and the tail, and the wound closes without a scar. That is remarkable, and it still happens in water that carries bacteria and fungus. Remove the cause, keep the water clean and cool, and watch the site daily for the signs on the infection page.

DO NOT MEDICATE A TANK ON A HUNCH

Amphibian skin absorbs what is in the water, and medicating the display tank can kill the filter's bacteria colony too. Correct the water first, and get a diagnosis before any medication goes in.

Tubbing, cooling & salt baths

Three interventions worth knowing before you need them, and where each one stops.

Tubbing

Tubbing means moving the axolotl out of its tank into a bare, unfiltered container of dechlorinated, temperature-matched water, changed completely rather than filtered. Research colonies house many adults this way as routine, in bowls with no circulating water, because water replaced regularly cannot build up waste. A tub is easy to watch against a plain background, isolates a sick or injured animal from a tankmate, and gives somewhere to go while a tank is put right.

- A clean plastic tote, bin or bucket the axolotl can turn around in without bumping the walls. No gravel and no sharp decor; one simple hide such as a length of PVC pipe.
- A full water change once daily as the baseline, twice for a genuinely sick animal, with fresh dechlorinated water checked by thermometer before the axolotl goes back in. A partial change defeats the point.
- Move the axolotl in a submerged container, never by hand.
- Keep testing the main tank in parallel: a tub fixes one animal's exposure, not whatever made the tank a problem.

Cooling as treatment

A sick axolotl on antibiotics for a bacterial infection may be held at 41 to 59°F (5 to 15°C) for several weeks, because cooler water appears to help. That belongs to a treatment a vet is running. Cooling a tub within the normal range as general support for a stressed animal is reasonable on your own, done gradually.

Salt baths

ELEMENT	THE PUBLISHED FIGURES
Strength	10 to 15 g/L
Time	No more than about 10 minutes
How often	Once or twice daily
What these figures are	The ceiling before salt itself starts damaging skin and gills, given as a caution against overdoing it. Not a prescription.
Not specified	Which condition a bath treats, or whether the salt should be iodized

If a vet prescribes a bath, expect the strength, the time and the type of salt to be set for that animal and that condition.

WHERE HOME CARE STOPS

A tub does not fix a bacterial infection, a spreading fungus, an impaction or an infected wound. Open sores or red streaking, a fungus not improving within a couple of days, uncontrollable floating, bloating, or appetite loss running several days all mean a vet. A tub can also hide subtle behavior changes, so it is a tool for a purpose, not a place to keep an axolotl just in case.

Setup checklist & targets

Tick through it before the axolotl comes home, and again whenever something seems off.

Equipment

- ☐ 20 gal long tank for one, 40 gal breeder for two (page 6)
- ☐ A secure lid
- ☐ Sponge filter, or a canister with a spray bar (page 10)
- ☐ A cooling plan: chiller, room air conditioning, or a fan and a cool room (page 8)
- ☐ A reliable thermometer in the water
- ☐ Liquid test kit: ammonia, nitrite, nitrate, pH
- ☐ Bare bottom, or fine sand under 1 mm (page 11)
- ☐ 2 or more smooth hides per axolotl, ideally 3 to 4, one a tunnel
- ☐ Dechlorinator for chlorine and chloramine
- ☐ Siphon and buckets used only for the tank
- ☐ Long, blunt feeding tongs
- ☐ Worms and a sinking pellet made for axolotls (page 14)

Targets

TARGET	RANGE
Water temperature	60 to 64°F (15.6 to 17.8°C). Never above 72°F (22°C); danger from about 75°F (24°C)
Ammonia and nitrite	0 ppm
Nitrate	Under 10 ppm
pH	7.4 to 7.6, with 6.5 to 8.0 tolerated if stable
Water change	20 to 30 percent weekly, dechlorinated and temperature-matched
Feeding	Adults every 2 to 3 days, juveniles about daily; leftovers out within 15 to 30 minutes
Light	Dim, on a timer for a 12-hour day

Before the axolotl comes home

- ☐ Tank cycled: a full ammonia dose reads 0 within 24 hours, nitrite 0, nitrate building (page 9)
- ☐ A water change done to bring nitrate under 10 ppm
- ☐ Temperature held in range through a warm day, and the cooling plan tested
- ☐ An amphibian or aquatic vet found, with the number on the emergency card
- ☐ Legal status confirmed for your state and city (page 22)

Emergency & quick targets

CALL AN AMPHIBIAN OR AQUATIC VET IF

Fungal growth not improving 48 hours after the water is corrected · bloating, floating, or no waste · red streaking, reddened skin or open wounds · a swollen, fluid-filled body · floating it cannot correct · appetite loss lasting 5 days or more · a known swallowed stone or piece of decor

READING	TARGET
Water temperature	60 to 64°F (15.6 to 17.8°C); never above 72°F (22°C)
Ammonia / nitrite	0 ppm / 0 ppm
Nitrate	Under 10 ppm
pH	7.4 to 7.6
Feeding	Adults every 2 to 3 days, juveniles about daily
Water change	20 to 30 percent weekly; at least a third at once if ammonia or nitrite shows
Salt bath, only if a vet directs	Published ceiling 10 to 15 g/L, no more than about 10 minutes, once or twice daily. Not a prescription

Your numbers

Vet _____

Tank volume _____

Emergency clinic that sees axolotls _____

Backup contact _____

Axolotl's name and color _____

Last vet visit _____

The water is too warm, right now

1. Lights off, lid cracked, fan across the water surface, blinds closed.
2. Float sealed bottles of frozen dechlorinated water and swap them as they melt. No loose ice.
3. No faster than about 1°F an hour. Test ammonia and nitrite while you wait.

HAVE THIS READY WHEN YOU CALL

The water temperature and how long it has been out of range, your latest ammonia, nitrite, nitrate and pH readings, a sample of tank water in a clean container, what it eats and when it last ate, and when the signs started. Transport in a clear, watertight container of its own tank water, kept cool (page 36).

Budget & shopping list

United States retail ranges, last checked September 2026. The rows add up to the totals.

INITIAL SETUP	TYPICAL COST
Tank: 20 gal long minimum, 40 gal breeder better	\$50 to \$150
Filtration: a low-flow sponge filter, or a canister rated for about 30 gal	\$20 to \$120
Substrate: fine sand under 1 mm (free if bare bottom)	\$10 to \$25
Hides, dechlorinator, thermometer and test kit	\$40 to \$80
Basic setup, without a chiller	\$120 to \$375
Aquarium chiller	\$150 to \$400
Setup with a chiller	\$270 to \$775

The axolotl itself is extra: \$30 to \$100 for a standard morph, more for a rare one such as a mosaic or GFP.

MONTHLY ONGOING	TYPICAL COST
Food, with nightcrawlers as the staple	\$5 to \$15
Water conditioner, filter media and chiller electricity	\$10 to \$30
Monthly ongoing	\$15 to \$45

VET FEES, ONE EXOTICS HOSPITAL'S 2026 LIST AS A BENCHMARK	COST
Standard wellness exam / medical exam	\$115 / \$135
Aquatic animal exam: 30 minutes with a technician, then 30 with the vet	\$235
Emergency exam, plus the after-hours fee	\$200 plus \$120

PLAN FOR, DON'T COUNT ON AVOIDING

Tests, medication and any surgery for a swallowed stone come after the exam fee. Keep a few hundred dollars set aside from the first week.

WHERE NOT TO CUT CORNERS

Cooling and the test kit. Warm water is the leading cause of trouble in this species, and the test kit is the only way to see the second one, bad water, coming. If the room cannot hold the water in range, the chiller is part of the real cost, not an upgrade.

First 30 days

Assumes the tank was cycled before the axolotl arrived.

Week 1

- ☐ Acclimate temperature first, then chemistry, then move it in a submerged container (page 17)
- ☐ Keep the light dim and leave it alone to settle
- ☐ Read the thermometer every day
- ☐ Test ammonia and nitrite 2 to 3 times this week, as in any new tank
- ☐ Offer food on the schedule for its age, and pull leftovers within 15 to 30 minutes

Week 2

- ☐ First weekly water change, on a fixed day, with nitrate tested before it
- ☐ Start the owner log: one row each test day (page 38)
- ☐ Check the filter flow is not pushing the axolotl around or curling its gills

Week 3

- ☐ Look at the gills in good light against how they looked in week 1
- ☐ Try tong feeding and a scattered meal (page 19)
- ☐ Call the vet you found and confirm they see axolotls, before you need them

Week 4

- ☐ Read four weeks of the log for trends, not single readings
- ☐ Siphon the floor and check nothing small enough to swallow has crept in
- ☐ Plan for summer: what holds the temperature now may not hold it in July (page 8)

Days 31 to 90

- ☐ Keep testing weekly, even when everything looks stable
- ☐ A newcomer to a home with another axolotl stays in quarantine 6 to 8 weeks (page 20)
- ☐ Fill in the pet-sitter sheet before you need it (page 37)

Symptom quick reference

Start every row by reading the thermometer and testing the water. It is the answer more often than not.

SYMPTOM	LIKELY CAUSE	ACTION
Gills curled forward	Heat, water quality or strong current	Check temperature, test, check the filter output (page 25)
White or gray cottony growth	Fungal infection	Fix the water and cool; vet if no better in 48 hours (page 26)
Red streaks, reddened skin, open sores	Bacterial infection	Vet the same day (page 24)
Red, inflamed patches on skin or gills	Ammonia or nitrite burn	Test, change at least a third, repeat until 0 (page 28)
Bloated and floating, no waste	Impaction	Stop feeding, call a vet (page 27)
Swollen, fluid-filled body	Ascites: heat or bacterial infection	Vet, promptly (page 26)
Floating, sluggish, in a warm tank	Heat stress	Cool gradually, about 1°F an hour (page 25)
Gulping air much more than usual	Not enough oxygen in the water	Check temperature and surface movement, add aeration (page 18)
Off food 5 days or more	Temperature, water, impaction or infection	Test first, then vet (page 24)
Eye changes	Prolonged high nitrate	Water changes to bring nitrate under 10 ppm, vet if it persists (page 28)
Adult getting heavy	Fed on a juvenile schedule	Every 2 to 3 days, head-sized portions (page 13)
Ammonia or nitrite above 0 ppm	Uncycled tank or a stalled filter	At least a third changed now, then retest daily (page 9)
Nitrate climbing past 10 ppm	Change overdue, or overfeeding	Larger or more frequent changes (page 12)
Water above 72°F (22°C)	Cooling not keeping up	Emergency cooling steps (page 31)
Missing toes or gill filaments	A tankmate biting	Separate them, keep the water clean (page 6)
Axolotl out of the tank	No secure lid	Back into its water at once, then secure the lid (page 6)

Daily, weekly & seasonal routine

About a minute a day, and one fixed day a week for the water.

Daily

- ☐ Read the water temperature
- ☐ A look at the gills in good light
- ☐ Filter running, surface moving
- ☐ Feed on schedule; leftovers out within 15 to 30 minutes

Weekly

- ☐ Test ammonia, nitrite, nitrate and pH, before the change
- ☐ Change 20 to 30 percent, dechlorinated and temperature-matched
- ☐ Siphon waste and leftovers off the floor
- ☐ Rinse the mechanical sponge or floss in the old water as it needs
- ☐ One row in the owner log

Monthly

- ☐ Rearrange one piece of non-critical decor, or rotate in one new smooth item
- ☐ Run a hand over the decor for new rough edges
- ☐ Read back through the log for slow trends
- ☐ Check the chiller or fan still does what it did last month

Seasonal

- ☐ Spring: test the cooling plan against a genuinely hot day, before summer
- ☐ Spring: have the fan, bottles for freezing and a battery air pump ready
- ☐ Summer: lights off on hot days, blinds closed
- ☐ Yearly: update the sitter sheet and confirm your vet still sees axolotls

FIX THE DAY, NOT THE INTENTION

Pick one day a week and always test and change water on it. A routine tied to a day survives a busy month. By the time an axolotl tank looks like it needs attention, the readings moved days ago.

WHAT IS NOT ON THIS LIST

Handling, full teardowns and medication. None belong in a routine. Biological filter media is rinsed barely ever, never all at once.

Power outages, heat waves, travel & transport

For an axolotl, a summer power outage is an oxygen problem and a heat problem at once.

When the power goes out

Oxygen is the fast risk. Once the filter stops, the surface stops moving, gas exchange slows, and dissolved oxygen starts falling within a couple of hours. So the first move is aeration, not the thermometer.

Do

- Run a battery air pump with an airstone. Own one before you need it.
- No pump: scoop water out and pour it back from a few inches up for a minute or two, every 10 to 15 minutes in a small tank, 20 to 30 in a larger one.
- Unplug the filter, so you decide when it restarts.
- In summer, keep heat out: blinds closed, the tank away from windows, the room as cool as you can.

Don't

- Don't feed until the filter is back and stable.
- Don't restart a filter that sat dead for hours without checking it: rinse the media in old tank water first, or decomposing waste flushes into the tank.
- Don't use loose ice. Sealed frozen bottles, swapped as they melt.
- Don't bring an open flame or a heat source near the tank.

Heat waves

The emergency runs in reverse for an axolotl: heat is the danger, not cold. Above about 75°F (24°C) it goes off food, develops ascites and floats. Use the cooling steps on page 25; for an animal already in crisis, a dish in the refrigerator is the recognized stopgap while the tank is cooled.

Travel

A sitter needs the pet-sitter sheet, walked through in person, and a thermometer they will actually read. In summer, the cooling has to work without you: test it before you leave.

Transport and moving house

- Move the axolotl in a small, clear, watertight container filled with its own tank water, kept cool.
- In warm weather: an insulated container in an air-conditioned car, never a hot trunk or a parked car.
- For a move, pack the filter media in a sealed bag or tub of tank water, never tap water, with a battery airstone on a long drive. Set the filter up again as soon as the tank is filled, using that reserved water.
- On arrival, match temperature before anything else, then move the animal in a submerged container.

FILL IN AND LEAVE BY THE TANK

Pet-sitter sheet

Everything a sitter needs on one page. Walk them through it once in person.

Axolotl's name and color _____

Feeding days _____

My phone _____

Backup contact _____

Vet _____

Where the bottles and fan are _____

THE ONE THING THAT MATTERS MOST

This animal must stay cold. The thermometer should read 60 to 64°F (15.6 to 17.8°C). Above 72°F (22°C) is an emergency: close the blinds, turn the light off, put the fan across the water, float a sealed bottle of frozen water, and call me. Never add a heater, and never put loose ice in the tank.

Each visit

- ☐ Read the thermometer and write the number down
- ☐ Check the filter is running and the water surface is moving
- ☐ Look at the axolotl: resting on the bottom, gills feathery and out to the sides, not curled forward
- ☐ Feed one pre-portioned meal, only on the days listed above
- ☐ Take out anything not eaten after 15 to 30 minutes

Please don't

- Feed extra or on another day. An adult eats every 2 to 3 days and always looks interested.
- Touch, lift or net the axolotl for any reason. Its skin and gills damage very easily.
- Add tap water that has not been dechlorinated, clean the filter, or add anything to the tank.
- Turn on a bright light over the tank. The axolotl has no eyelids.

CALL ME IF

The thermometer reads above 72°F (22°C) · the filter is silent · the axolotl is floating and cannot get down · there is white fluff, a red patch or a sore on it · the power has been out more than an hour or two. If I don't answer, call the vet.

Owner log

One row a week, filled in on test day. Photocopy this page, or track digitally. Consistency matters more than the format.

DATE	TEMP	AMMONIA	NITRITE	NITRATE	PH	% CHANGED	APPETITE, GILLS, NOTES

WHY BOTHER LOGGING

The water readings are this animal's vital signs, and they take minutes with no handling at all. An axolotl often shows a water problem in its gills or appetite, and the log shows which reading moved first. Sixteen rows is four months, long enough to see a real trend, and it is the most useful thing you can hand a vet.

Equipment & vet log

Equipment log

ITEM	INSTALLED	REPLACE OR CHECK BY	DONE
Tank: volume and date set up			
Filter, and how the output is spread			
Biological media last rinsed			
Chiller or fan: model and setting			
Test kit opened, and its expiry date			
Battery air pump and spare batteries			

Twelve-month planner

Tick the box in the month you did it. Month 1 is whichever month you start in.

TASK, BY MONTH	1	2	3	4	5	6	7	8	9	10	11	12
Rearrange or rotate decor	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Check chiller or fan	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Read back the log	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Test the cooling plan	☐			☐			☐			☐		
Battery pump test	☐					☐						☐

Vet visit log

DATE	REASON AND READINGS	WHAT WAS DONE OR PRESCRIBED	OUTCOME

Record every salt bath, medication and water emergency here too. A vet will ask.

Enrichment checklist & log

Setup

- ☐ 3 to 4 hides of different shapes, one a tunnel
- ☐ A still corner away from the filter output
- ☐ Dim light on a timer, with plant cover or floating plants for shade
- ☐ Feeding tongs, and a sand saucer if you use one

Monthly review

- ☐ Rearrange one piece of non-critical decor
- ☐ Rotate one new smooth item in for 5 to 7 days
- ☐ Vary feeding: tongs, scatter, a different spot every few days
- ☐ Every item still smooth, stable, and too big to swallow (page 11)

Enrichment log

DATE	WHAT YOU TRIED	HOW IT WENT, AND WHETHER TO REPEAT

ENRICHMENT IS NOT EXTRA FOOD

Every foraging idea here uses the meal the axolotl was already getting, presented differently. An adult still eats every 2 to 3 days, whether the worm arrives on tongs, scattered, or hidden in a sand saucer.

REFERENCE

Glossary, A to L

Ammonia	The first waste product of an axolotl and of rotting food, and highly toxic. The target reading is 0 ppm. Page 9.
Ascites	Fluid swelling in the abdomen. A sign of heat stress above about 75°F (24°C) or of a bacterial infection, and a vet visit. Page 26.
Baffle	Anything that breaks up a filter's outflow, such as foam or a spray bar, so the current reaching the axolotl is gentle. Page 10.
Bd	<i>Batrachochytrium dendrobatidis</i> : a chytrid fungus that causes the skin disease chytridiomycosis in more than 200 amphibian species. Page 20.
Bsal	<i>Batrachochytrium salamandrivorans</i> : a chytrid fungus that kills salamanders, axolotls included. The reason for the 2025 federal listing of genus <i>Ambystoma</i> . Pages 20, 23.
Captive-bred	Hatched from parents kept by people rather than caught in the wild. Pet axolotls are captive-bred. Page 4.
Chiller	A device that actively cools aquarium water. The only cooling method that holds a set temperature whatever the weather. Page 8.
Chloramine	A compound of chlorine and ammonia added to many city water supplies. Unlike chlorine it does not leave water that is left standing, so a dechlorinator that handles it is needed. Page 9.
Chytrid fungi	A group of fungi that infect amphibian skin. Bd and Bsal are the two that matter for pets. Page 20.
Cloaca	The single vent behind the back legs, used for waste and breeding. Swollen in a mature male. Page 21.
Cycling	Growing the filter bacteria that turn ammonia into nitrite and nitrite into nitrate. Four to six weeks is typical, slower in cold water. Page 9.
Dechlorinator	A water conditioner that neutralizes chlorine and chloramine in tap water before it reaches the axolotl. Page 9.
Drip-matching	Adding small amounts of the destination water to a new arrival's container over time, after the temperatures match, so the chemistry shifts gradually. Page 17.
Fecal test	A vet's microscope check of a fresh stool sample for parasites, best done during quarantine. Pages 20, 28.
GFP	Green fluorescent protein: a protein from a jellyfish gene that glows green under ultraviolet light. Put into axolotls for research in 2006, and now a pet morph. Page 5.
GH	General hardness: the amount of dissolved minerals in water. No published target exists for axolotls. Page 9.
Impaction	A gut blockage from something swallowed that cannot pass, most often gravel or small decor. Page 27.
Injurious wildlife	Animals the federal government lists under the Lacey Act as harmful, which limits importing and shipping them. Page 23.
IUCN	International Union for Conservation of Nature: the body that keeps the Red List of threatened species, which grades the axolotl Critically Endangered. Page 4.
KH	Carbonate hardness: the water's buffering capacity, which steadies pH. No published target exists for axolotls. Page 9.
Lacey Act	The federal wildlife law under which genus <i>Ambystoma</i> was listed as injurious wildlife in 2025. Page 23.
Lateral line	A row of pressure-sensing organs along the body that lets an axolotl feel movement in the water, including food. Page 18.
Leucistic	The mostly white morph with dark eyes. Not the same as albino, which has no black pigment at all. Page 5.

Glossary, M to Z

Melanoid	A solid dark morph, with many black pigment cells and none of the reflective ones. Page 5.
Morph	A color variety. Morphs differ in looks only, never in care. Page 5.
Mosaic and chimera	An axolotl built from two genetically different cell lines, carrying patches of two colors. It does not breed true. Page 5.
Nitrate	The end product of the nitrogen cycle. Far less toxic than ammonia or nitrite, kept under 10 ppm by water changes. Page 9.
Nitrite	The middle product of the nitrogen cycle, and toxic. The target reading is 0 ppm. Page 9.
Nitrogen cycle	The chain in which filter bacteria turn ammonia into nitrite and then nitrate. Page 9.
Paedomorphosis	Reaching adult size and breeding while keeping the larval body: gills, a finned tail and a fully aquatic life. Page 5.
pH	How acidic or alkaline water is, on a scale where 7 is neutral. 7.4 to 7.6 for an axolotl. Page 9.
ppm	Parts per million: how many parts of a substance there are in a million parts of water. The unit of a water test reading. Page 9.
PVC	Polyvinyl chloride: a rigid plastic. Wide pipe sections with sanded edges make safe hides. Pages 11, 32.
Quarantine	Keeping a new axolotl in its own tank or tub, with its own tools, for 6 to 8 weeks before it meets an established one. Page 20.
Regeneration	Regrowing lost body parts. Axolotls regrow limbs, the tail, and parts of the eye, brain, spinal cord and heart. Page 28.
RO	Reverse osmosis: a filtering process that strips almost everything out of water. Never used straight for an axolotl without remineralizing. Page 9.
Salt bath	A short bath in salted water. The published figures, 10 to 15 g/L for about 10 minutes, are a ceiling, so a vet sets the dose. Page 29.
Sand saucer	A shallow dish of sand under 1 mm with food hidden in it, for searching without loose sand across the floor. Page 14.
Spray bar	A bar with a row of small holes on a filter or chiller return, which spreads the flow out. Page 10.
Sponge filter	An air-driven filter that pulls water through a foam block. Gentle by design, the standard choice for axolotls. Page 10.
Tubbing	Keeping an axolotl in a bare, unfiltered container with a full daily water change, for isolation, observation or support during illness. Page 29.
Un-ionized ammonia	The more toxic of ammonia's two forms. Its share rises with temperature and pH. Page 7.
Units	°F, degrees Fahrenheit, and °C, degrees Celsius, for temperature; in (inches), cm (centimeters) and mm (millimeters) for size; gal (US gallons) and L (liters) for tank volume; mL (milliliters) and g (grams) for small amounts; g/L (grams per liter) for a salt bath; ppm for water tests.
VCA	Veterinary Centers of America, now VCA Animal Hospitals: an animal hospital chain whose pet care articles are among this package's sources.
Water mold	The white, cotton-like fungal growth that is the most common fungal infection in axolotls. Page 26.

REFERENCE

Sources

Where the numbers in this package come from, so you can check them yourself.

Veterinary references

- LafeberVet: Basic Information Sheet, Axolotl. Water targets, size, maturity, cohousing, feeding, obesity, parasites, lighting and gloves.
- R. Loh, World Small Animal Veterinary Association 2015 World Congress Proceedings: Common Disease Conditions in Axolotls, and Water Quality Explained. Heat signs and ascites, the refrigerator stopgap, septicemia and its treatment, and the salt bath ceiling.
- VCA Animal Hospitals: Axolotls as Pets. Chicago Exotics Animal Hospital and Tree of Life Exotic Pet Medical Center: axolotl care. Portions, handling, nets and buying checks.
- Merck Veterinary Manual: Environment and Husbandry for Amphibians, and Environmental Diseases of Aquatic Animals. The 6 to 8 week quarantine, separate tools, the bleach dilution, and 1°F an hour.
- Association of Reptile and Amphibian Veterinarians: the find-a-vet directory. Colorado Exotic Animal Hospital: exam pricing.

Research and husbandry

- Ambystoma Genetic Stock Center, University of Kentucky: Guide to Axolotl Husbandry, and Axolotl Strains. Colony temperatures, bowl housing, mineralized water, behavior, cleaning, larval growth and color strains.
- Servín, Alcantar-Rodriguez and Medrano (2025), *Frontiers in Amphibian and Reptile Science*: tongs, pellet protein and breeding after a cool spell. Billingham (2017), *Animal Technology and Welfare*: axolotl enrichment and tunnels.
- Page and colleagues (2008), *BMC Genomics*: thyroid hormone and metamorphosis. Joven, Elewa and Simon (2019), *Development*: regeneration. Sobkow and colleagues (2006), *Developmental Biology*: the GFP line. Woodcock and colleagues (2017), *Scientific Reports*; Kabangu and colleagues (2023), *Genes*; Houillon and Bagnara (1996), *Pigment Cell Research*: the morph genes and chimeras.
- San Diego Zoo Wildlife Alliance Library: axolotl reproduction. Association of Zoos and Aquariums, Amphibian Taxon Advisory Group: quarantine. National Research Council: laboratory amphibian guidelines.
- US Geological Survey, National Wildlife Health Center: Bsal and Bd. University of Florida Institute of Food and Agricultural Sciences Extension and the US Environmental Protection Agency: ammonia and oxygen. Texas A&M AgriLife Extension: temperature swings. A chiller maker's manual: rated volumes.
- IUCN Red List: *Ambystoma mexicanum*. Conservation International (2025): the Xochimilco survey counts and threats.

The law

- California Code of Regulations, title 14, § 671; New Jersey Administrative Code 7:25-4.10 and the Division of Fish and Wildlife restricted species notice; New Mexico Director's Species Importation List; Wyoming Game and Fish Commission Chapter 69; Alabama Administrative Code r. 220-2-.26; District of Columbia Code § 8-1808; Maine Unrestricted Species List; Vermont Restricted Wild Animal List; Code of Massachusetts Regulations, title 321, § 9.01; West Virginia Code § 20-2-13; Hawaii List of Conditionally Approved Animals; Rhode Island exotic animal rules § 3.7; Minnesota Statutes § 97A.401; Arkansas unrestricted captive wildlife list; and each other state's rules, read on 1 October 2026. US Fish and Wildlife Service: salamanders as injurious wildlife, and its answers on the 2025 rule. Virginia Department of Wildlife Resources: the 2021 change.

The site this package is built on

- Beastly Facts guides, current at publication: the seven axolotl guides, the encyclopedia entry and profile article, cold and clean water, and the shared guides to cycling, water changes, filtration, outages and transport, cooling, amphibian quarantine, and tubbing and salt baths. Prices last checked September 2026.

REFERENCE

Where the sources disagree

Where credible sources give different numbers, this package prints one and says why.

TOPIC	WHAT THE SOURCES SAY	WHAT THIS PACKAGE USES
Temperature	60 to 64°F (a veterinary information sheet); 60 to 65°F, never above about 72°F (a research colony)	60 to 64°F, with 72°F as the ceiling every source agrees on
Adult size	Most 9 to 10 in (a veterinary sheet); 9 to 12 in (the species profile)	9 to 12 in
Maturity	About one year (a research colony); males about 10 months, females 12 to 18 months (veterinary sources)	Males about 10 months, females 12 to 18 months
Meal size	Roughly the size of the head (veterinary sources); as much as it eats in 5 to 10 minutes (an exotics hospital)	Both: a head-sized portion, what it takes in 5 to 10 minutes
Weekly water change	About 30 percent (a veterinary sheet); 20 to 30 percent (care guides); far smaller volumes changed every other day (research colonies)	20 to 30 percent weekly in a cycled tank, more if nitrate climbs
Nets	A soft net or a container (a veterinary clinic); a container, with a soft net only briefly when nothing else works (amphibian guidance)	A container; a soft net for seconds only as a last resort
Quarantine	6 to 8 weeks (veterinary guidance); 30 days minimum, 60 preferred (zoos)	6 to 8 weeks, restarted if illness appears
Hardness	Specific GH and KH bands repeated widely, with no primary source found behind them	No number. Never straight RO or distilled water
Salt baths	Figures repeated as a treatment dose; the veterinary source gives 10 to 15 g/L for 10 minutes as the ceiling before harm	The ceiling only, with the dose set by a vet
Chiller	Essential (several care sources); a fan and a cool room suffice in cooler climates (others)	Decided by the room: a chiller once it runs above the mid 60s°F for weeks

Version history & about

What changed in each edition, so a printed copy can be placed.

Version history

This is edition 3.0 (Oct 2026). The store page shows the current edition; re-download free if yours is older. Where this list and a page disagree, the page wins.

- **3.0, Oct 2026.** Rebuilt on the current outline, 45 pages. Safety: target water temperature 60 to 64°F (15.6 to 17.8°C); worms for adults cut into head-sized pieces, never whole; a meal is what the axolotl takes in 5 to 10 minutes; any ammonia or nitrite reading means changing at least a third of the water, 50 percent or more if it is climbing; a fishless cycle dosed to 3 ppm and finished when a full dose reads 0 within 24 hours; if an axolotl must be touched, clean disposable nitrile gloves; a soft net for seconds only when nothing else works; tools disinfected with bleach at 30 mL per liter for 30 minutes, then rinsed and dried; maturity at about 10 months for males and 12 to 18 for females; adult size 9 to 12 in; the antibiotic dose figure removed, since dosing is the vet's. Legal: two pages on the law in all 52 jurisdictions, with West Virginia added as permit-only and New Jersey's basis rewritten to its named ban. Costs: setup \$120 to \$375, or \$270 to \$775 with a chiller. Other changes are wording only.
- **2.2, Sep 2026.** Legal: every state re-read against its own rule; the District of Columbia and New Mexico moved to banned; Wyoming, Alabama, Vermont, Massachusetts, Hawaii, Rhode Island, Minnesota and Arkansas added. Other changes are wording only.
- **2.1, Sep 2026.** Safety: salt bath figures of 10 to 15 g/L for 10 minutes shown as a published ceiling, not a dose; temperature changed no faster than 1°F an hour; the hardness target withdrawn; quarantine 6 to 8 weeks; ascites and the refrigerator stopgap added. Other changes are wording only.
- **2.0, Sep 2026.** Rebuilt on the current template, 41 pages.
- **1.0, Aug 2026.** First edition, 20 pages. Replaced by 2.0.

This package is a husbandry reference compiled from the listed sources and cross-checked against current veterinary guidance at the time of publication. It is not a substitute for an amphibian or aquatic veterinarian, and nothing in it should delay a vet visit. Treatments described here are for recognizing what a vet may do, never for use at home. The law pages are a summary as of October 2026, not legal advice.

© Beastly Facts · Edition 3.0 · October 2026 · For personal use by the purchaser. Please don't redistribute.